

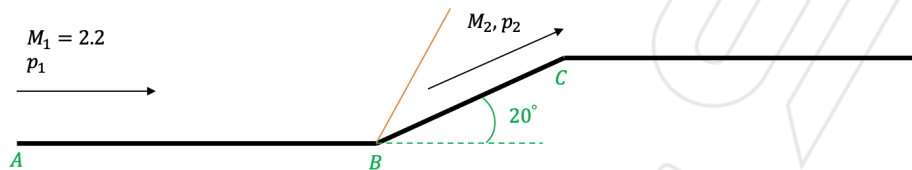
AERO50001 Aerodynamics 2 Problem Set #6

Tutorial 2

6.1 Oblique shock in ramp

★☆☆ 5-15 mins

The following figure shows an oblique shock forming when air ($\gamma = 1.4$) flowing at $M_1 = 2.2$ over a wall finds a corner with deflection of 20° .



- Use the oblique shock tables to find the wave angle of the oblique shock shown in the figure (at least two decimal spaces).
- Find the Mach number M_2 behind the shock wave.
- Find the pressure on the wall BC if the upstream pressure is atmospheric.
- What is the minimum upstream Mach number for which the flow deflection can be achieved with a straight oblique shock? Give your answer to at least two decimal places.

a) $\beta = 47.976^\circ$

b) $M_2 = 1.404$

c) $p_2 = \frac{p_2}{p_1} p_1 = 2.949 \cdot 1 \text{ atm} = 2.949 \text{ atm}$

d) If $\theta_{\max} = 20^\circ$, $\beta = 64^\circ \quad \therefore M = 1.85$

6.2 Oblique and normal shock combinations

★★☆ 10-15 mins

We want to decelerate air at a Mach number of 3.2 and at sea level conditions. We consider two cases:

- the flow passes through a single normal shock wave
 - the flow first goes through an oblique shock wave with $\beta = 30^\circ$, followed by a normal shock.
- (a) Calculate the stagnation pressure behind the shock in the first case (a single normal shock wave).
- (b) Calculate the stagnation pressure behind the shock in the second case (an oblique shock wave with $\beta = 30^\circ$, followed by a normal shock).
- (c) Which shock wave arrangement is more efficient in decelerating supersonic flow (fewer losses, that is, smaller drop in stagnation pressure)?
- The single normal shock
 - The oblique shock followed by a normal shock

$$(a) \quad M_2^2 = \frac{2 + (\gamma - 1) M_1^2}{2\gamma M_1^2 - (\gamma - 1)} \quad \therefore M_2 = 0.4643$$

$$\therefore \frac{p_2}{p_1} = \frac{2\gamma M_1^2}{\gamma + 1} - \frac{\gamma - 1}{\gamma + 1} = 11.78$$

$$\frac{p_{02}}{p_2} = \left(1 + \frac{\gamma - 1}{2} M_2^2\right)^{\frac{\gamma}{\gamma - 1}} = 1.159$$

$$\therefore p_{02} = \frac{p_{02}}{p_2} \frac{p_2}{p_1} p_1 = 13.66 p_1 = 13.66 \times 101325 \text{ Pa} = 1.3836 \text{ MPa}$$

$$(b) \quad \text{for } \beta = 30^\circ, \quad M_1 = 3.2 \quad \therefore \theta = 14^\circ$$

$$\therefore M_2^2 \sin^2(\beta - \theta) = \frac{2 + (\gamma - 1) M_1^2 \sin^2 \beta}{2\gamma M_1^2 \sin \beta - (\gamma - 1)}$$

$$\therefore M_2 = 2.453 \quad \therefore M_3 = 0.5144$$

$$\therefore p_{03} = \frac{p_{03}}{p_{02}} \frac{p_{02}}{p_1} p_1 = \frac{p_{03}}{p_{02}} \frac{p_{02}}{p_2} \frac{p_2}{p_1} p_1 = 2.34 \text{ MPa}$$

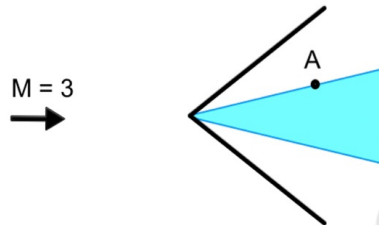
(c) second one

6.3 Supersonic flow past a wedge

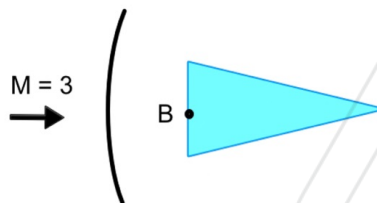
★★☆ 8-15 mins

Consider air flowing supersonically at $M = 3$ and with pressure $p = 70$ kPa towards a double symmetric wedge of 16° (total angle) at zero incidence. The wedge can be oriented pointed end forward or blunt end forward.

(a) If the pointed edge is facing forward, what will be the pressure at point A?



(b) If the blunt edge is facing forward instead, what will be the pressure at point B?



(a) $M_1 = 3, \theta = 8^\circ \therefore \frac{p_2}{p_1} = 1.795 \therefore p_2 = 1.795 p_1 = 125.65 \text{ kPa}$

(b) At B, just behind central normal shock wave section:

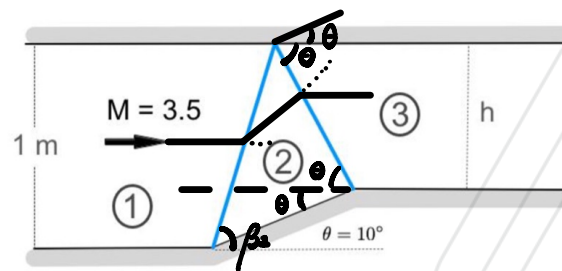
$$M_1 = 3, M_3 = 0 \therefore M_2 = 0.4752$$

$$\frac{p_2}{p_1} = \frac{p_{02}}{p_1} = 12.06 \therefore p_2 = 12.06 p_1 = 844.2 \text{ kPa}$$

6.4 Inlet geometry

★★★ 20-30 mins

For a specific design Mach number M , the geometry of an inlet can make the reflected shock wave cancel. Consider the geometry shown in the figure below (not to scale) for a supersonic diffuser to reduce the Mach number from $M = 3.5$. The deflection angle is $\theta = 10^\circ$.



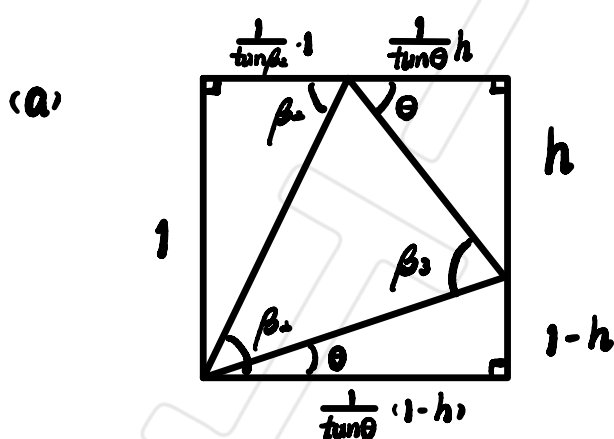
- Find the downstream height h .
- Find the downstream Mach number M_3 .

(b) $M_1 = 3.5, \theta = 10^\circ$

$$\therefore \beta_2 = 24.304^\circ \quad \therefore M_2 = 2.904$$

for $\theta = 10^\circ, M_2 = 2.904$

$$\therefore \beta_3 = 28.130^\circ \quad \therefore M_3 = 2.423 \quad \therefore \theta = 18.130^\circ$$



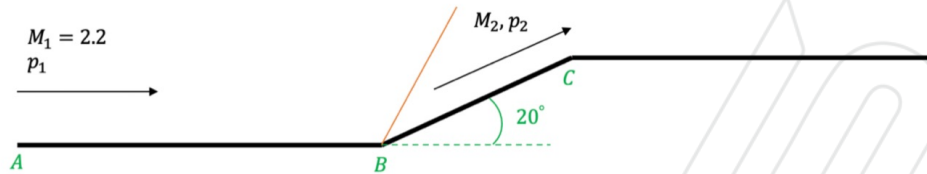
$$\therefore \frac{1}{\tan \theta} (1-h) = \frac{1}{\tan \beta_2} 1 + \frac{1}{\tan \theta} h$$

$$\therefore h = \frac{1 \cdot \left(\frac{1}{\tan \theta} - \frac{1}{\tan \beta_2} \right)}{\left(\frac{1}{\tan \theta} + \frac{1}{\tan \theta} \right)} = 0.397 \text{ (m)}$$

6.5 Prandtl-Meyer expansion

★★☆ 10-20 mins

The flow downstream of the shock wave in the figure below is turned through a Prandtl-Meyer expansion at C, until it has returned to its original flow direction.



- (a) Determine the Mach number downstream of the expansion.
- (b) Determine the pressure downstream of the expansion, if upstream the pressure is atmospheric.

$$(a) \quad M_1 = 2.2, \theta_1 = 20^\circ$$

$$\therefore M_2 = 1.404$$

$$\therefore \theta_1 + \nu(M_2) = \theta_2 + \nu(M_3)$$

$$\therefore 20^\circ + 9.106^\circ = 0^\circ + \nu(M_3)$$

$$\therefore \nu(M_3) = 29.106^\circ \quad \therefore M_3 = 2.1$$

$$(b) \quad \frac{p_2}{p_1} = 2.949$$

$$\frac{p_2}{p_1} = \frac{p_2}{p_2} \frac{p_2}{p_1} = \frac{p_2}{p_2} \frac{p_2}{p_2} \frac{p_2}{p_2} \frac{p_2}{p_1}$$

$$= \frac{1}{9.145} \cdot 1 \cdot 3.200 \cdot 2.949 = 1.032$$

$$\therefore p_2 = 1.032 p_1 = 1.032 \text{ atm}$$